Diabetes Risk Prediction using a Lightweight SQLite ETL and Random Forest Pipeline



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1. Introduction

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Motivation

- Diabetes is a growing global health crisis.
- Early detection reduces complications and healthcare costs.

Problem Statement

- Many small clinics lack large-scale data infrastructure.
- Clinical data is often unstructured (CSV, missing values).
- Need for a lightweight yet effective ML pipeline.

Objective

Build an end-to-end diabetes prediction pipeline:

- Based on SQLite for ETL
- Uses Random Forest for modeling
- Visualizations and UI for clinical use



2. Related Work

Existing Studies

1. Hasan & Yasmin (2025)
    - Compared classifiers (RF, SVM, XGBoost, CNN+LSTM)
    - Found Random Forest robust with tuning
2. Probst et al. (2018)
    - Hyperparameter strategies for RF
    - Suggested grid search and model-based optimization
3. El Massari et al. (2024)
    - Used RFECV + SMOTE
    - Showed feature selection boosts accuracy



Our Contribution

- Integrate ETL + EDA + RF modeling in a single SQLite-backed Python pipeline
- Use real-world healthcare features
- Include a Streamlit app for clinical predictions

3. Methodology

Diabetes prediction dataset

gender	age	hypertension	heart_disease	smoking_history
Gender refers to the biological sex of the individual, which can have an impact on their susceptibility to diabetes.	Age is an important factor as diabetes is more commonly diagnosed in older adults.Age ranges from	Hypertension is a medical condition in which the blood pressure in the arteries is persistently elevated. It	Heart disease is another medical condition that is associated with an increased risk of developing diabetes. It	Smoking history is also considered a risk factor for diabetes and can exacerbate the complications associated

bmi	HbA1c_level	blood_glucose_le...	diabetes
BMI (Body Mass Index) is a measure of body fat based on weight and height. Higher BMI values are linked to a	HbA1c (Hemoglobin A1c) level is a measure of a person's average blood sugar level over the past 2-3 months. Higher	Blood glucose level refers to the amount of glucose in the bloodstream at a given time. High blood glucose	Diabetes is the target variable being predicted, with values of 1 indicating the presence of diabetes and 0



ETL Pipeline

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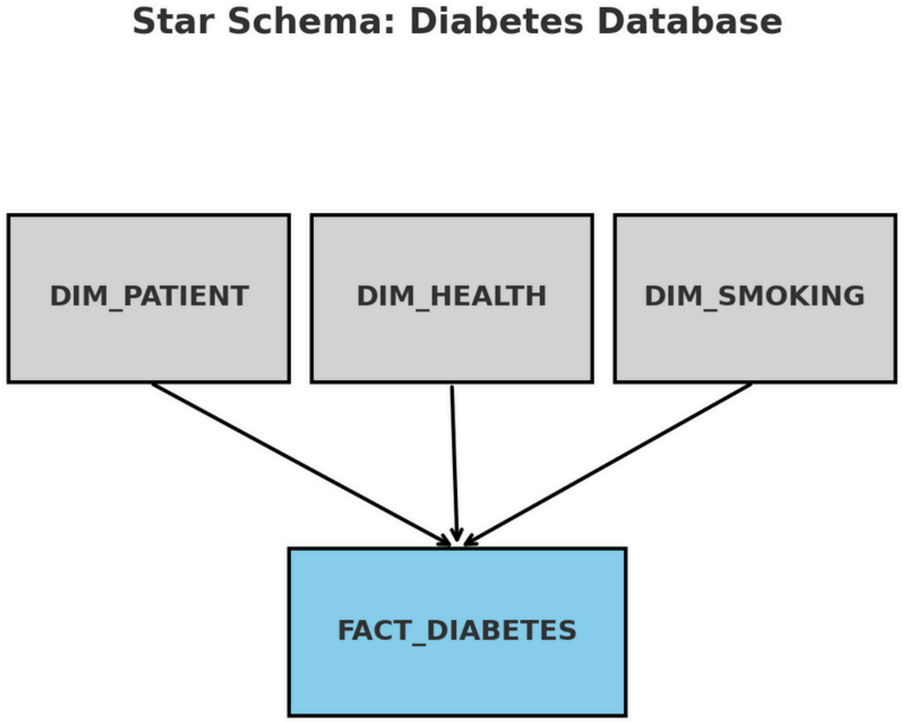
Stage	Description	Tools/Techniques Used
Extract	- Load raw CSV using pandas - Inspect headers and values - Detect missing or placeholder values (e.g., 0 in HbA1c, BMI)	pandas
Transform	- Replace invalid/zero values with **median imputation** - Normalize data into **Star Schema** (Fact + Dimensions) - One-hot encode categories	pandas, numpy, star schema design
Load	- Insert cleaned data into **SQLite** database - Define tables: FACT_DIABETES, DIM_PATIENT, DIM_HEALTH, DIM_SMOKING - Use foreign keys	sqlite3, SQL schema design
Outcome	- Structured, queryable clinical dataset - Ready for SQL-based EDA and ML modeling - Lightweight, reproducible, and scalable architecture	Easily extendable in low-resource environments

Star Schema

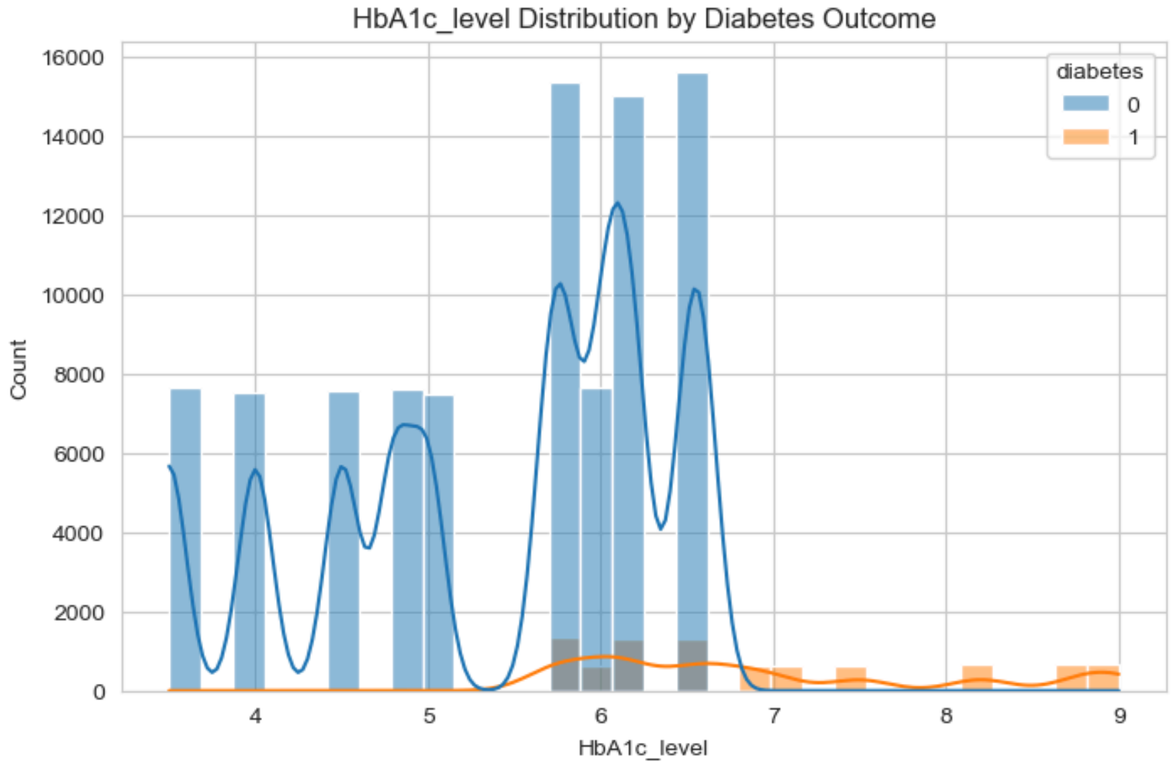
FACT DIABETES: Contains numeric measures (BMI, HbA1c, blood glucose level) and the target label (diabetes). References all dimension tables via foreign keys.

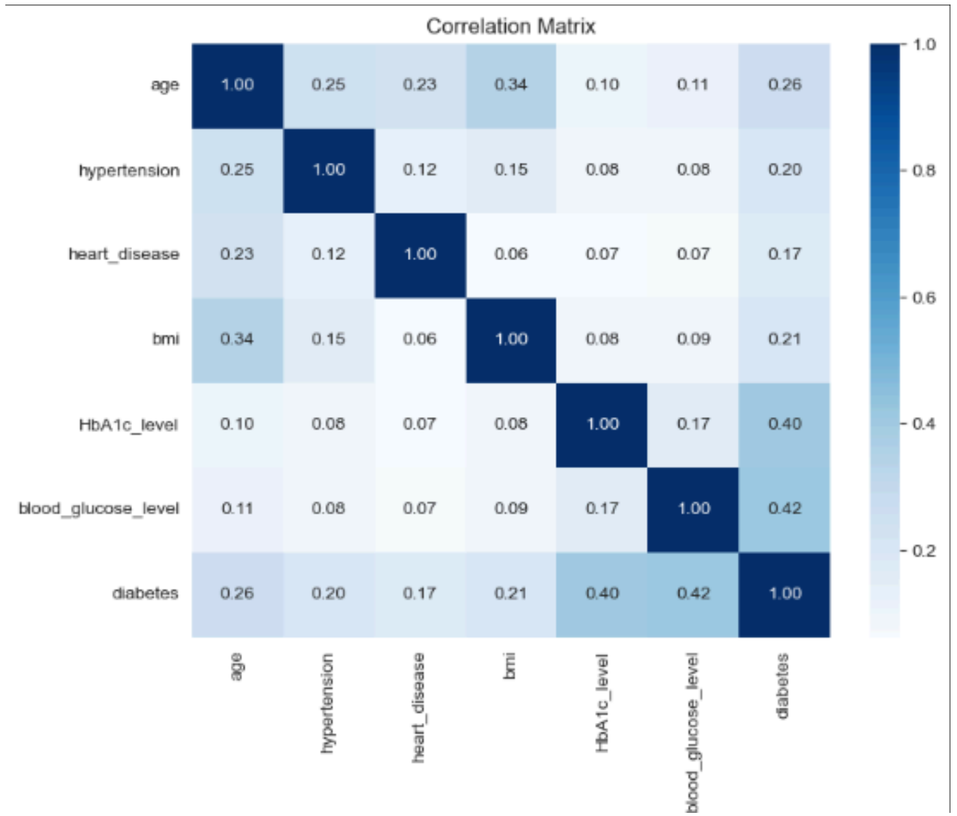
- DIM PATIENT: Stores patient attributes like gender and age.
- DIM HEALTH: Stores binary indicators such as hypertension and heart disease.

DIM SMOKING: Encodes smoking history categories (e.g., never, current, former).



Exploratory Data Analysis (EDA)

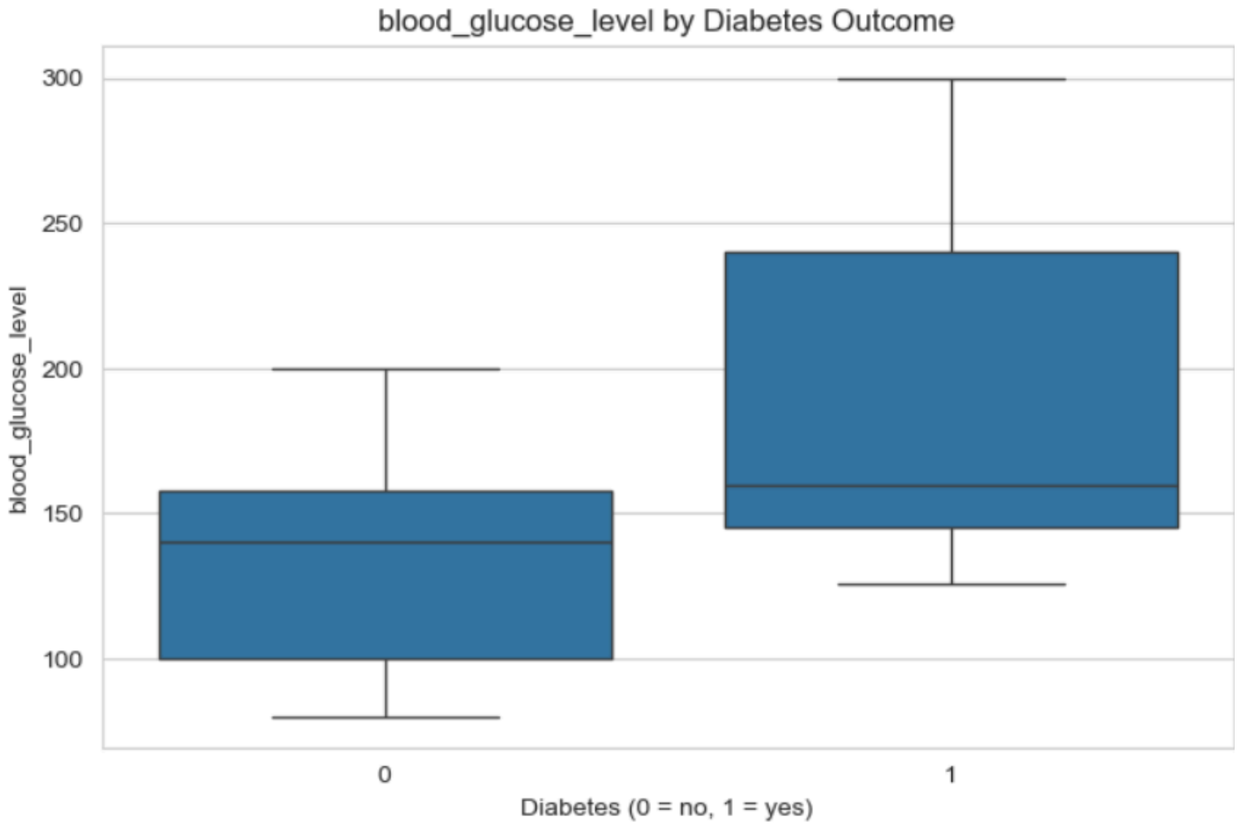




- HbA1c (Hemoglobin A1c) reflects average blood glucose over 2–3 months.
- Higher values typically indicate poor glucose control and are associated with diabetes.

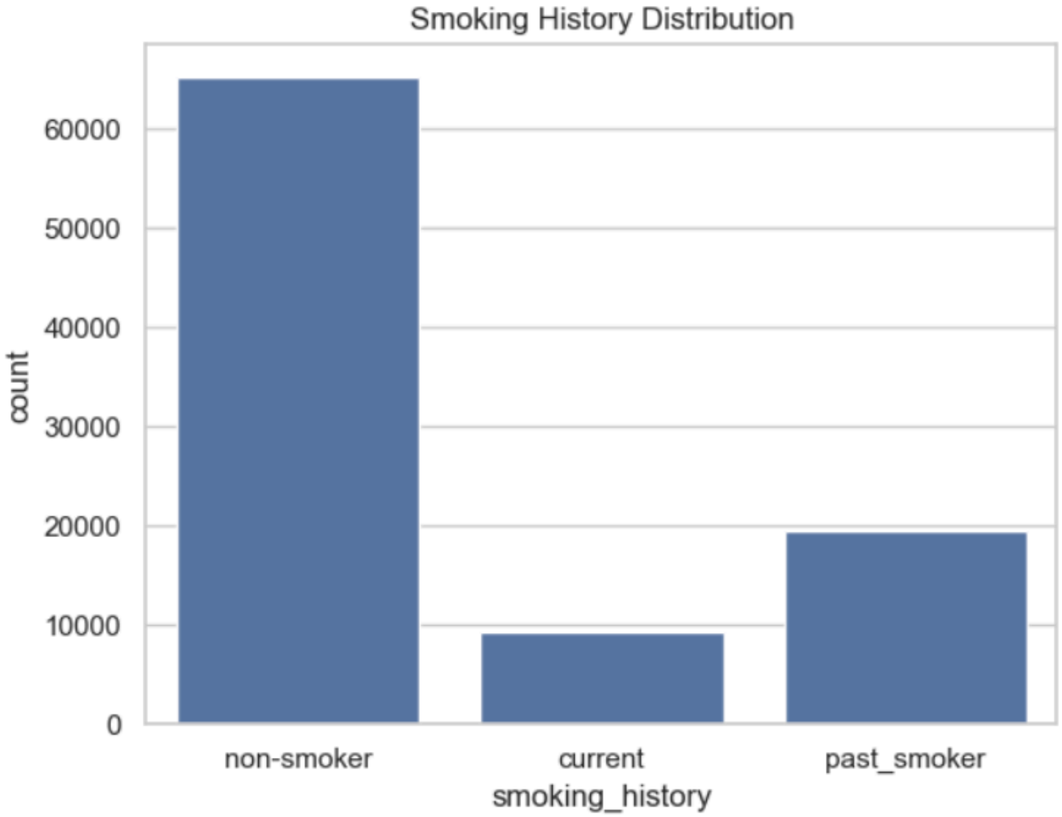
blood_glucose_level and HbA1c_level

Exploratory Data Analysis (EDA)



distribution of blood glucose levels for two groups based on diabetes status

Diabetic patients tend to have higher blood glucose levels overall.



shows that the data is skewed (or unbalanced)

Model Training and Evaluation

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Baseline Performance: Initially, the model was trained using default hyperparameters on an 80:20 train-test split. Performance was assessed using accuracy, precision, recall, F1-score, and confusion matrix. As shown in Table I, the baseline model achieved solid predictive results, with an accuracy of 85%

BASELINE RANDOM FOREST PERFORMANCE

Metric	Value
Accuracy	0.85
Precision	0.83
Recall	0.81
F1-score	0.82

Hyperparameter Tuning:

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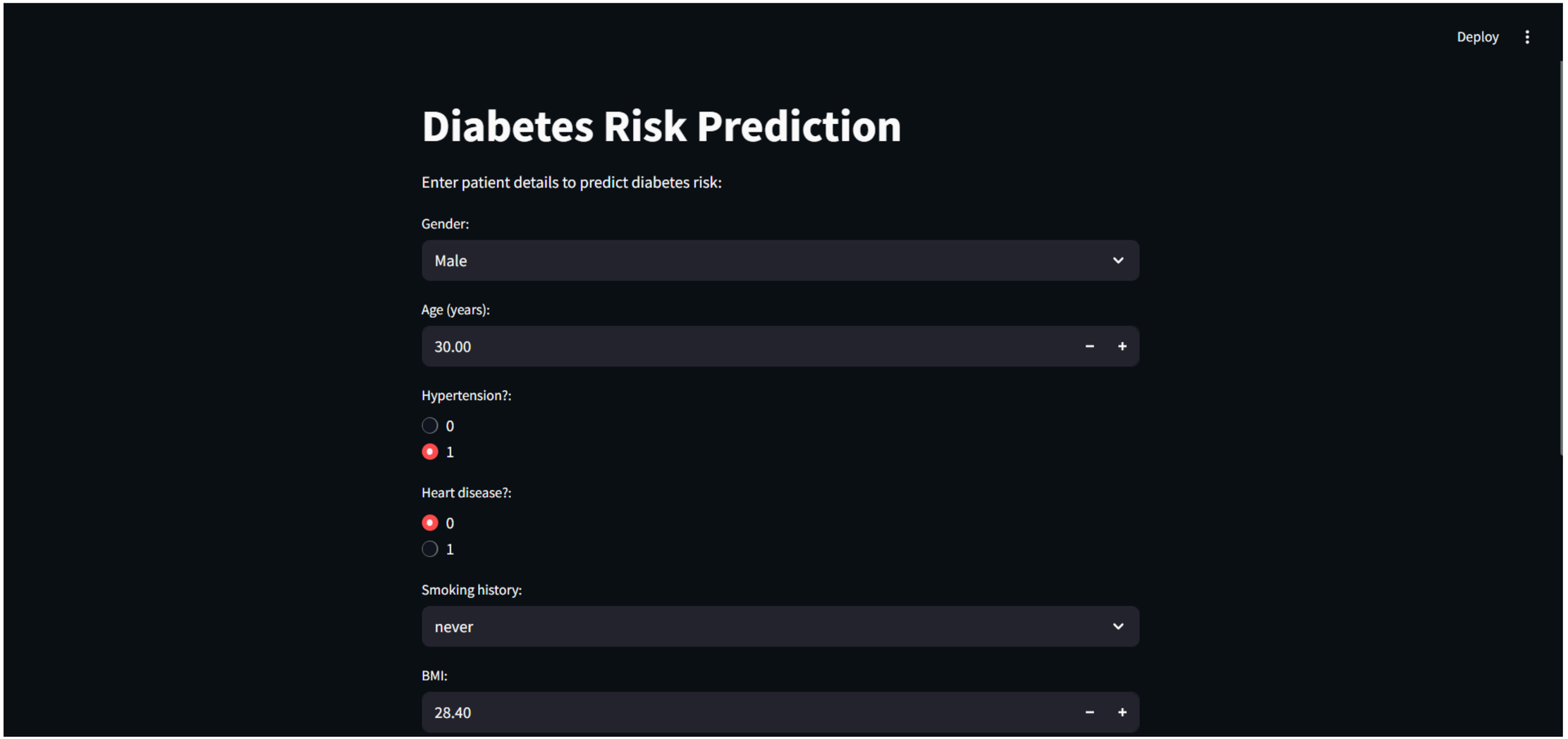
To improve generalization and reduce variance, a hyperparameter optimization step was conducted using GridSearchCV. The grid search explored combinations of the number of estimators (n_estimators), maximum depth of trees (max_depth), and the minimum number of samples required to split an internal node (min_samples_split).

After tuning, the model's accuracy increased to 95%, with marked improvements in both recall and F1-score, indicating better handling of class imbalance.

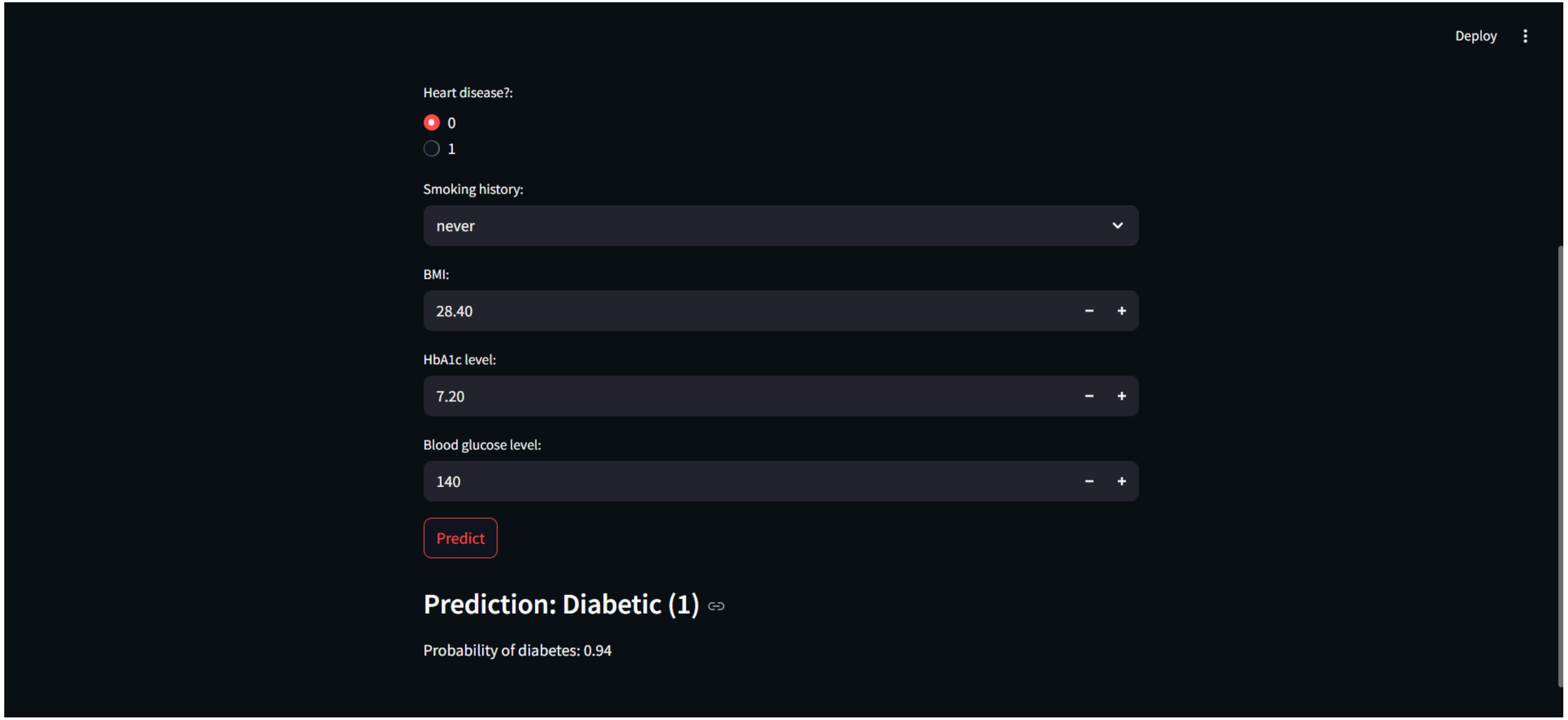
TABLE II
MODEL PERFORMANCE: BASELINE VS TUNED

Metric	Baseline RF	Tuned RF
Accuracy	0.85	0.95
Precision	0.83	0.96
Recall	0.81	0.95
F1-score	0.82	0.95

Streamlit Dashboard

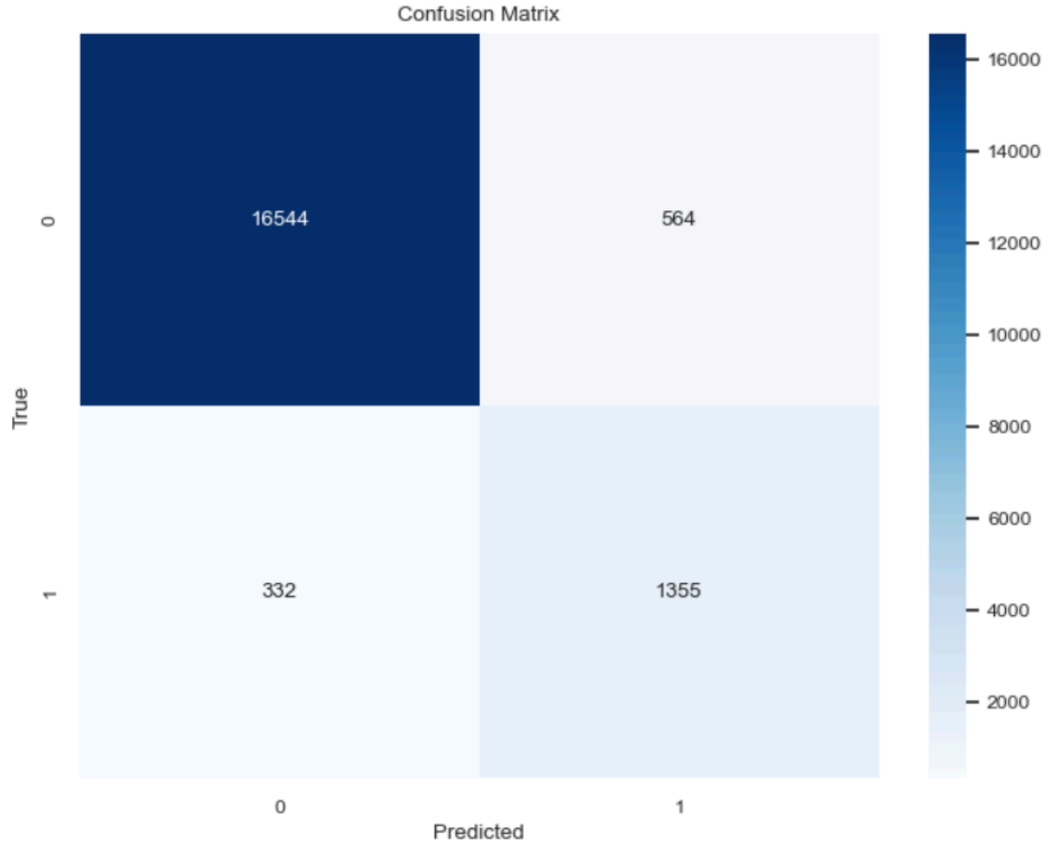


Streamlit Dashboard



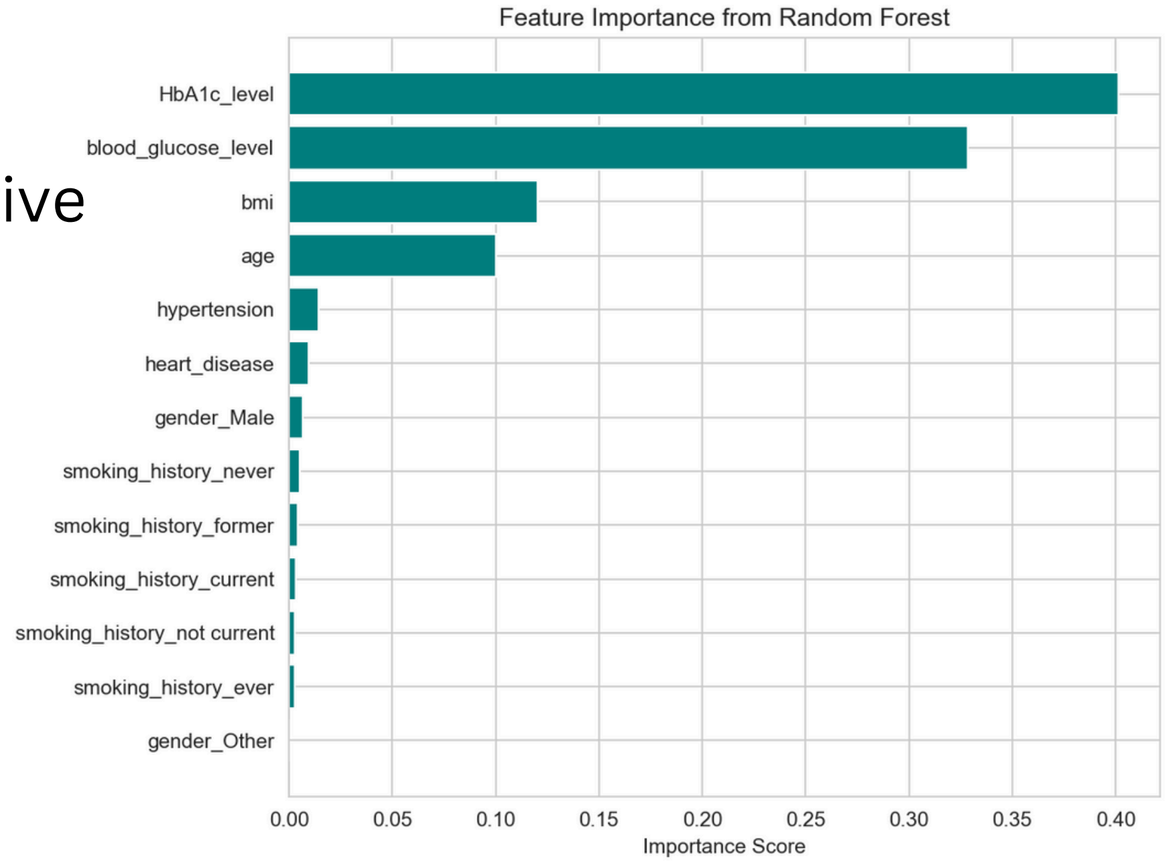
Confusion Matrix Analysis

The classifier demonstrates
a strong ability to correctly identify both diabetic and non-diabetic cases, with minimal false negatives—an essential quality in medical risk prediction systems.



Feature Importance and Interpretation

The top features ranked by their relative importance. The model identified blood_glucose_level, HbA1c_level, and BMI as the most predictive features—aligning with established medical research that highlights these markers as key indicators of metabolic health and diabetes risk



Conclusion

- Developed an efficient and reproducible diabetes prediction system.
- Integrated ETL, EDA, ML, and UI in a single pipeline.
- Random Forest + SQLite is accurate, lightweight, and interpretable.
- Achieved high predictive power (95%) using domain-relevant features.



Future Work

Integration of Additional Clinical Variables: Including lab tests, family history, or longitudinal data may improve model accuracy and generalizability.

• Explainability Techniques: Implementing SHAP or LIME explanations could increase interpretability and clinical trust.

• Model Comparison: Additional ensemble methods like XGBoost and deep learning models could be compared using the same pipeline to validate Random Forest's effectiveness.

• Data Warehouse Extension: Migrating from SQLite to a cloud-based data warehouse (e.g., BigQuery, Snowflake) may support real-time data updates and larger-scale inference.

